

1. Which of the following types of applications are most suitable for an Online Transactional Processing (OLTP) system ?

- Student Information System such as eVision
- Data warehouse
- AI model training
- Image processing

Ans : Student Information System such as eVision

2. What command could be used to find several values in the text field of the weather collection:

- `db.weather.find({text:{$all:["sun","rain","icy"]}})`
- `db.weather.find({text:{$in:[/sun/,/rain/,/icy/]}})`
- `db.weather.find({text:"sun rain icy"})`
- `db.weather.find({$or:["sun","rain","icy"]})`

Ans : `db.weather.find({ text: { $in: [/sun/, /rain/, /icy/] }})`

3. Given 1 Kilobyte is 1000 bytes. How many multiples of kilobytes is an Exabyte (EB)

- 1000^3
- 1000^4
- 1000^5
- 1000^6

Ans : 1000^6

4. What does PageRank calculate ?

- The authority of a web page
- The speed at which a web page loads in a browser
- The number of images and multimedia files present on a web page
- The total amount of text content available on a web page

Ans : The authority of a web page.

5. What sort of data is best suited to a bitmap index:

- High-cardinality data
- Low-cardinality data (gender, yes/no)
- Text data
- Video data

Ans : Low-cardinality columns such as gender

6. Consider the following query vector q and document vectors d_1 and d_2 . According to the vector space model, what would be the scores of d_1 and d_2 and how would the documents be ranked ?

$$\vec{q} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$\vec{d}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad \vec{d}_2 = \begin{pmatrix} 1 \\ 0.5 \\ 1 \end{pmatrix}$$

- d2 with score 2
d1 with score 1
- d1 with score 2
d2 with score 1
- d1 with score 1
d2 with score 1
- d2 with score 3
d1 with score 2

Ans :

- d2 with score 2
d1 with score 1

7. One approach to handling dirty data is to Fix it. What would this entail?

- Delete dataset
- Replace incorrect values with correct values
- Ignore errors
- Duplicate data

Ans : Replace the incorrect value with the correct value

8. Using a Spark DataFrame and the Weather json file, which of the following commands would show just the screen name and name of tweets made by German users:

- `df.select('user.screen_name','user.name').show(40)`
- `df.filter(df['user.lang'] == "de").select('user.screen_name','user.name').show(40)`
- `df.filter(df['user.name'] == "de").show(40)`
- `df.groupBy('user.lang').count().show(40)`

Ans : `df.filter(df['user.lang'] == "de") .select('user.screen_name', 'user.name').show(40)`

9. What is the purpose of stopword elimination ?

- Frequent terms like “and”, “or”, don’t bear any meaning for search and retrieval. This means we can eliminate them to keep our vocabulary small and improve the efficiency of text processing and indexing in natural language applications.
- The goal is to increase the amount of textual data by adding commonly used words so that machine learning models can capture more linguistic variations and improve generalization across different datasets.
- It is used to translate all words in a dataset into multiple languages so that search systems can support multilingual queries and improve accessibility for users across different regions and languages.
- It helps in converting raw text documents into structured image-based representations so that computer vision models can be used for analyzing textual information instead of traditional NLP methods.

Ans : Frequent terms like “and”, “or”, don’t bear any meaning for search and retrieval. This means we can eliminate them to keep our vocabulary small.

10. Which of the following is not a valid command to handle data in HDFS?

- `hdfs dfs -put file.txt /user/data/`
- `hdfs dfs -ls /user/data/`
- `hdfs dfs -rm /user/data/file.txt`
- `cp -r /user/data /user/test/`

Ans : `cp -r /user/data /user/test/`

11. Which of the following is an example of photo and video image data:

- Text file
- Spreadsheet
- Digital image
- Database table

Ans : Digital image

12. Which type of model is used for predicting a continuous numerical value?

- Regression Model
- Classification Model
- Clustering Model
- Reinforcement Model

Ans : Regression Model

13. Name the high level language that is a main part of Apache Pig

- Pig Script
- Pig Latin
- Pig SQL
- Pig Query

Ans : Pig Latin

14. If today's temperature is 12oc .What type of measurement is the temperature Value?

- Nominal
- Ordinal
- Interval
- Ratio

Ans : Interval

15. Given the following data, what type of NoSQL database would suitable:

- Key-value database
- Document database
- Column-family database
- Graph database

EMPNO	ENAME	MGR	HIREDATE	SAL	DEPTNO
7876	ADAMS	7788	19-Nov-15	1100	20

EMPNO	ENAME	MGR	HIREDATE	SAL	COMM	DEPTNO
7499	ALLEN	7698	20-Feb-95	1600	300	30

EMPNO	ENAME	HIREDATE	SAL
7839	KING	17-Nov-80	5000

EMPNO	ENAME	MGR	HIREDATE	SAL	DEPTNO	EMAIL
7788	SCOTT	7566	16-Oct-15	3000	20	scott.tiger@oracle.com

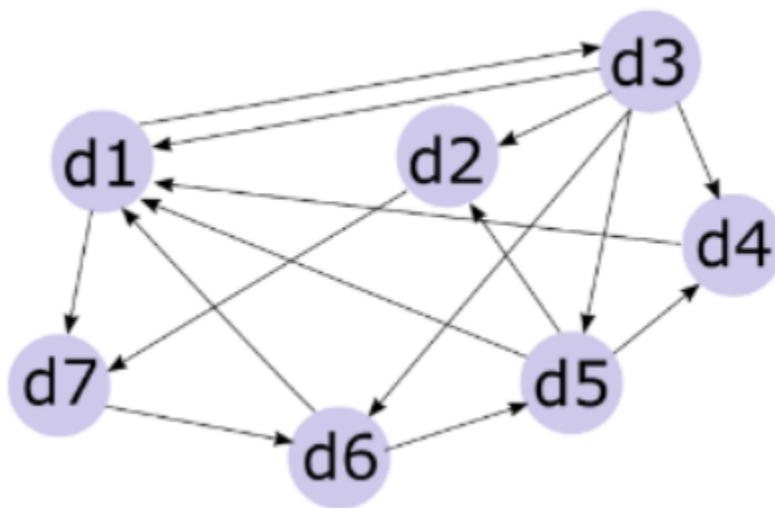
Ans : Column-family database

16. Which of the following requirements are needed for programming Big Data:

- Handle fault tolerance
- Access data fast
- Distribution computations to nodes
- All of the answers

Ans : All of the answers

17. Below is the mini toy Web, which of the web pages would you expect to get the highest PageRank ?



Ans : d1

18. What does Veracity mean in the context of Big Data:

- Speed of data generation
- Trustworthiness of data
- Size of data
- Storage format

Ans : Refers to the trustworthiness of the data

19. You have a new collection called myTweets which has over 10,000 documents.

You do not know what type of data it contains. Which command can help you find out the structure of a document?

- db.myTweets.find()
- db.myTweets.findOne()
- db.myTweets.schema()
- db.myTweets.show()

Ans : db.myTweets.findOne()

20. What is the order of the three steps to Map Reduce?

- Map → Reduce → Shuffle
- Reduce → Map → Sort
- Map → Shuffle and Sort → Reduce
- Sort → Map → Reduce

Ans : Map -> Shuffle and Sort -> Reduce

21. The following is a snippet of Java code from the main method for the Word Count program:

```
public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "word count");
    job.setJarByClass(WordCount.class);
    job.setMapperClass(TokenMapper.class);
    job.setCombinerClass(IntSumReducer.class);
    job.setReducerClass(IntSumReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
} // main
```

Which line of code tells Hadoop what output directory to create?

Ans : FileOutputFormat.setOutputPath(job, new Path(args[1]));

22. What command can be used to find documents containing the word icy in the

weather collection. icy should be found no matter what case it is in (upper, lower, etc).

- db.weather.find({text:"icy"})
- db.weather.find({text:/icy/i})
- db.weather.search("icy")
- db.weather.findCase("icy")

Ans : db.weather.find({text: /icy/i})

23. What is the fundamental unit of data in Apache Spark?

- DataFrame
- RDD
- Dataset
- Table

Ans : RDD

24. Given the below inverted index, which documents would be retrieved for the query

retrieval systems OR multimedia OR images

retrieval systems

DOCID	POS
d1	10,25,37
d2	23
d4	45,55
d7	3,56

multimedia

DOCID	POS
d4	18,20
d6	22,27
d7	5
d9	3,96,127

images

DOCID	POS
d1	14,68
d3	29,73,235
d4	11,145
d7	10

- d1, d2, d3, d4, d6, d7, d9
- d2, d3, d5, d8
- d1, d5, d6, d8, d10

- d3, d4, d5, d7, d9

Ans : d1, d2, d3, d4, d6, d7, d9

25. What does the term NoSQL represent?

- Not Structured Query Language
- Not Only SQL
- New Object SQL
- None of the above

Ans : Not Only SQL

26. Given a collection called myModules, what would be the MongoDB command to list all the data in this collection:

- db.myModules.show()
- db.myModules.find()
- db.find(myModules)
- db.myModules.list()

Ans : db.myModules.find()

27. What is the purpose of lemmatization in NLP?

- Convert text into numerical format
- Reduce words to their root form
- Remove punctuation marks
- Identify part of speech tags

Ans : Reduce words to their root form

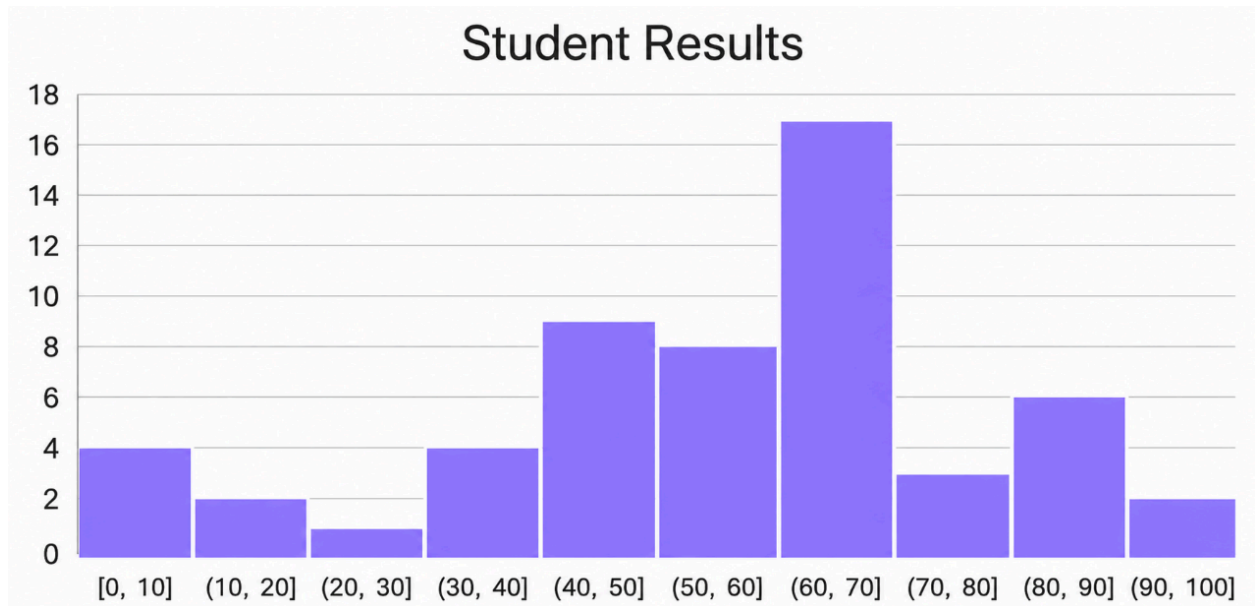
28. Which Python library is commonly used for EDA?

- Scikit-learn
- TensorFlow
- Matplotlib
- PyTorch

Ans : Matplotlib

29. What type of graph is this an example of:

- Bar chart
- Histogram
- Pie chart
- Line graph



Ans : Histogram

30. How does the HITS algorithm compute ?

- Hub and authority values for web pages
- Only the number of incoming links to a webpage for ranking purposes
- The loading speed of web pages based on server response time
- The total word count of a webpage to determine its importance

Ans : Hub and authority values for web pages.

31. Which of the following best explains the function of a Secondary NameNode in traditional HDFS architecture?

- It stores all replicated blocks when DataNodes fail
- It acts as a live standby NameNode automatically
- It assists by merging fsimage and edit logs periodically
- It replaces clients during read operations

Ans: It assists by merging fsimage and edit logs periodically

32. Sqoop is used for:

- Querying logs
- Data transfer between RDBMS and Hadoop
- Monitoring cluster
- Security scanning

Ans: Data transfer between RDBMS and Hadoop

33. YARN stands for:

- Yet Another Resource Negotiator
- Your Application Resource Node
- Yellow Area Resource Network
- None

Ans: Yet Another Resource Negotiator

34. In the command `reduceByKey(lambda x, y: x + y)`, what operation is actually being performed?

- Words are removed if repeated
- Counts belonging to the same word are added together
- Keys are converted into values
- Words are split into characters

Ans: Counts belonging to the same word are added together

35. A correlation coefficient value of 0 generally suggests that:

- There is no linear relationship between the variables
- Variables are perfectly positively related
- Variables are perfectly negatively related
- One variable causes the other

Ans: There is no linear relationship between the variables

36. In the following aggregation snippet:

```
$filter: {  
  input: "$employees",  
  as: "employee",  
  cond: { $gte: ["$$employee.sal", 2000] }  
}
```

What is the correct interpretation of `$$employee.sal`?

- It refers to a global variable named employee
- It refers to a field inside the current document being processed in the array
- It refers to the collection name employees
- It refers to the output of \$project stage

Ans: It refers to a field inside the current document being processed in the array

37. What is the main purpose of an inverted index in search engines?

- To store images and videos efficiently
- To map documents to words they contain
- To compress web pages for faster loading
- To encrypt search engine queries

Ans: To map documents to words they contain

38. In MongoDB, which statement best describes the nature of documents and ObjectIds?

- MongoDB requires a fixed schema using CREATE TABLE before inserting any document, and ObjectId must always be user-defined like a primary key.
- MongoDB is schema-less, meaning documents in the same collection can have different structures, and it automatically generates a unique ObjectId unless manually provided.
- MongoDB enforces a strict schema like relational databases, but ObjectId is always system-generated and cannot be modified.
- MongoDB stores data only in relational format, and ObjectId is used only for indexing, not uniqueness.

Ans: MongoDB is schema-less, meaning documents in the same collection can have different structures, and it automatically generates a unique ObjectId unless manually provided.

39. In MongoDB's aggregation framework, what is the correct purpose of using an aggregation pipeline such as \$project, \$filter, and \$size together?

- It permanently modifies the original documents in the collection before storing them
- It is used only to sort documents and does not allow transformations or calculations
- It processes data in multiple stages to transform, filter, and compute results without changing the original collection
- It replaces the need for all queries like `find()` and works only on single documents

Ans: It processes data in multiple stages to transform, filter, and compute results without changing the original collection.

40. Which of the following statements best describes the role of components in the Hadoop ecosystem?

- HDFS is used for data processing, MapReduce is used for storage, and Hive is used for hardware management
- HDFS stores large datasets across clusters, MapReduce processes the data in parallel, and Hive provides a SQL-like interface for querying data

- YARN is responsible for storing data blocks, while HBase is used for scheduling jobs in Hadoop
- Pig is used for cluster management, and Sqoop is used for real-time streaming of data

Ans: HDFS stores large datasets across clusters, MapReduce processes the data in parallel, and Hive provides a SQL-like interface for querying data

41. In MongoDB, which of the following is the correct way to store a date value in a document so that it is treated as a proper Date data type rather than a string?

```
"hiredate": "1989-06-09"
"hiredate": Date("1989-06-09")
"hiredate": new Date("1989-06-09")
"hiredate": (1989,06,09)
```

Ans: "hiredate": new Date("1989-06-09")

42. Which of the following correctly describes the basic working flow of a web search engine?

- Query is directly matched with databases → results are manually filtered → pages are stored
- Web crawler collects pages → indexer organizes content → query processor retrieves ranked results
- Users send queries → servers generate websites → crawler deletes irrelevant pages
- Index is created first → then web pages are manually written → results are cached permanently

Ans: Web crawler collects pages → indexer organizes content → query processor retrieves ranked results

43. In the MapReduce programming model, what is the primary function of the Mapper phase?

- It merges all results from different nodes into a final output
- It stores data permanently in HDFS after processing
- It processes input data and generates intermediate key–value pairs
- It schedules jobs across the Hadoop cluster

Ans: It processes input data and generates intermediate key–value pairs

44. MongoDB stores data internally in which format?

- XML documents stored in tables
- BSON documents inside collections
- CSV files inside databases
- Plain JSON files without modification

Ans: BSON documents inside collections

45. HBase performs best under which condition?

- When used on a single machine with no cluster
- When running on less than 3 DataNodes
- When deployed on at least 6–10 nodes including HDFS DataNodes and a NameNode
- When used only for small datasets under 1 GB

Ans: When deployed on at least 6–10 nodes including HDFS DataNodes and a NameNode

46. Elasticsearch is built on which core concept for organizing and searching data?

- Relational tables and SQL joins
- Indexes containing documents for fast search retrieval
- File blocks stored in HDFS
- Key-value pairs stored only in memory

Ans: Indexes containing documents for fast search retrieval

47. What is the main responsibility of the ApplicationMaster in YARN?

- Store HDFS data blocks permanently
- Manage cluster-wide resource allocation
- Coordinate execution of a specific application by requesting resources and tracking tasks
- Replace the ResourceManager in case of failure

Ans: Coordinate execution of a specific application by requesting resources and tracking tasks

48. What is the primary purpose of Hadoop in big data systems?

- To design user interfaces for web applications
- To store and process large datasets in a distributed environment
- To replace all relational databases
- To execute only real-time transactions

Ans: To store and process large datasets in a distributed environment

49. According to Eric Schmidt in 2010, how much data is generated every two days:

- 500 Terabyte (TB)
- 5 Exabyte (EB)
- 50 Petabyte (PB)
- 1 Zettabyte (ZB)

Ans: 5 Exabyte (EB)

50. Which of the following types of applications are most suitable for a Big Data system:

- Personal note-taking app
- Massive Grid Computer System such as CERN's Large Hadron Collider Computing Grid
- Basic calculator software
- Offline text editor

Ans: Massive Grid Computer System such as CERN's Large Hadron Collider Computing Grid